Summary

- A caching layer can speed up your application significantly, while also lowering the costs of your primary data store.
- To keep the cache in sync with the database, items usually expire after some time, or a write-through strategy is used.
- When the cache is full, the least frequently used items are usually evicted.
- ElastiCache can run Memcached or Redis clusters for you. Depending on the engine, different features are available. Memcached and Redis are open source, but AWS added engine-level enhancements.
- Memcached is a simple key-value store allowing you to scale available memory and throughput by adding additional machines—called shards—to the cluster.
- Redis is a more advanced in-memory database that supports complex data structures like sorted sets.
- MemoryDB is an alternative to ElastiCache and provides an in-memory database with strong persistence guarantees. MemoryDB comes with Redis compatibility.
- CloudWatch provides insights into the utilization and performance of ElastiCache.